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This is a document that I am preparing as I read about various bend-and-break theorems. Let us fix the notational convention that we are going to use in these notes. All schemes are of finite type over a field  $k$ . By a variety we mean a separated, integral scheme of finite type over a field. A curve is a 1-dimensional connected, reduced proper scheme over a field  $k$ , not necessarily irreducible. By a curve on a variety  $X$  we mean a nonconstant morphism from a curve with smooth irreducible components to  $X$ , or equivalently its image in  $X$ .

First let me mention a few things about morphisms that contract curves. By the famous Stein Factorization theorem, any proper morphism  $f : X \rightarrow Y$  between varieties can be factorised as  $X \xrightarrow{f'} Y' \xrightarrow{g} Y$  where  $f'$  has connected fibers and  $g$  is a finite morphism. Actually having connected fibers is a consequence of the fact that in the proof of Stein Factorisation we take  $Y' = \mathbf{Spec}(f_*\mathcal{O}_X)$  which implies that  $\mathcal{O}_{Y'} \cong f'_*\mathcal{O}_X$ . It is important to note that the curves contracted by  $f$  and  $f'$  are the same.

**Remark:** If  $f$  has connected fibers to begin with then  $g$  is bijective. Let  $k$  be a field of characteristic 0. Then by generic smoothness theorem, there exists a dense open subset  $V \subseteq Y$  such that  $g|_{g^{-1}(V)} : g^{-1}(V) \rightarrow V$  is smooth. Since  $g|_{g^{-1}(V)}$  is flat and  $g$  is bijective,  $g|_{g^{-1}(V)}$  is a bijective finite étale morphism, which implies that it is an isomorphism. Hence  $g$  is a birational morphism. Moreover if  $Y$  is normal then for any open subset  $U \subseteq Y$ ,  $\Gamma(g^{-1}(U), \mathcal{O}_{Y'}) \cong \Gamma(U, \mathcal{O}_Y)$  since  $g$  is finite and  $\Gamma(U, \mathcal{O}_Y)$  in its fraction field. Hence  $g$  is an isomorphism.